FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Advanced)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CRT–III
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	D	B
2.	C	B	A
3.	D	A	A
4.	A	C	B
5.	B	A	B
6.	B	C	C
7.	B	A	B
8.	C	A	D
9.	B, C	A, C, D	B, C
10.	A, B	C, D	B, C, D
11.	C, D	B, C, D	A, B, C
12.	A, C, D	A, B, C, D	A, D
13.	C	C	A
14.	C	B	C
15.	A	C	A
16.	A	B	A
17.	D	A	B
18.	C	B	C
1.	2	3	1
2.	1	4	1
3.	4	3	5
4.	2	3	2
5.	5	8	2

Physics

PART – I

SECTION – A

1. $\Delta\phi = \frac{2\pi}{\lambda}\Delta X$

$\Delta\phi = \frac{2\pi}{16} \times 4 = \frac{\pi}{2}$

$I = 4I_0\cos^2\frac{\pi}{4} = 2I_0$

2. $mV_0R = \frac{3}{2}mR^2\omega$

$\omega = \frac{2V_0}{3R}$

$V_{cm} = R\omega = \frac{2V_0}{3}$

3. Electric field lines kill off on surface of conductor

4. Torque will be in clockwise sense.

5. Since maximum current passing through two circuits is same and equal to V/R_1 in (b) and V/R_2 in (c) then $R_1 = R_2$
Current growth equation of L – R circuit is :

$i = i_0\left(1 - e^{-t/\tau}\right)$

$i = i_0\left(1 - e^{\frac{-tR}{L}}\right)$

Now, $\frac{di}{dt} = \frac{i_0R}{L}e^{-tR/L}$

At t = 0 slope of I vs t curve:

$\frac{di}{dt} = \frac{i_0R}{L}$

From graph (a) : slope of curve (b) > slope of curve (c)

$\frac{R_1}{L_1} > \frac{R_2}{L_2}$

Since, $R_1 = R_2$

$L_1 < L_2$

6. Momentum conservation.

$MV_0 = 2MV$

$V = \frac{V_0}{3}$

Angular momentum conservation about centre of mass,

$\frac{1}{2}MR^2 \cdot \omega_0 - \frac{4}{3}MV_0R = \left[\frac{MR^2}{2} + M\left(\frac{4}{3}R\right)^2 + \frac{1}{2}(2M)R^2 + 2M\left(\frac{2}{3}R^2\right)\right]\omega$

$\omega = \frac{3}{25}\omega_0 - \frac{8}{25}\frac{V_0}{R}$

For $\omega = 0$

$$\omega_0 = \frac{8V_0}{3R}$$

The energy loss,

$$\Delta E = E_i - E_f$$

$$= \frac{1}{2}MV_0^2 + \frac{1}{2}\left(\frac{MR^2}{2}\right)\cdot\omega^2 - \frac{1}{2}(3M)V^2 = \frac{19}{9}MV_0^2$$

7. $$V_1 = \sqrt{\frac{T}{\mu_1}} = 10\sqrt{2}$$

$$V_2 = \sqrt{\frac{T}{\mu_2}} = \frac{10\sqrt{2}}{\sqrt{3}}$$

Reflection coefficient

$$R = \frac{V_2 - V_1}{V_1 + V_2} = 2 - \sqrt{3}$$

Transmission coefficient,

$$T = \frac{2V_2}{V_1 + V_2} = \sqrt{3} - 1$$

8. Initially,

$$P_{gas} = P_0A + mg = 112.5 \text{ kPa}$$

$$w = \int_0^x P_{gas}A\, dx$$

$$= \int_0^x (P_0A + mg + kx)dx = 56.5 \text{ J}$$

$$Q = m\Delta u + w = 118 \text{ J}$$

9. When the switch 'S' is opened,

$$\tau = RC = 1.5\,\text{sec}$$

Voltage across the capacitor,

$$V_c = 10\left(1 - e^{-t/1.5}\right) \text{ for } t < T$$

The current, $i = \frac{dQ}{dt} = C\ \frac{dV_c}{dt}$

$$= \frac{200}{3}\times 10^{-6}\left(e^{-t/1.5}\right) \text{ for } t < T$$

Voltage drop across 100 k μ resister,

$$V = iR = \frac{20}{3}e^{-t/1.5} \text{ for } t < T$$

After t = T,

$$\tau = 100\ \text{k}\Omega \times 10\ \mu\text{F} = 1\ \text{sec}$$

Voltage across capacitor,

$$V_c = V_o t^{-[(t-T)/\tau]} = 10\left(1 - e^{-(T/1.5)}\right), e^{-[(t-T)/1]}$$

The charge on capacitor,

$$Q = V_c \cdot C$$

The current through 100 k μ resistor,

$$i = -C\frac{dV_c}{dt} = 10^{-3}\left(1 - e^{-T/1.5}\right)\left(1 - e^{-(t-T)/1}\right)$$

10. If $\frac{\partial}{\partial y}\left(\frac{\partial V}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial y}\right)$ then $\bar{E}$ exists.

11. For emission of photoelectrons, $\lambda_{incident} < 5200\,\overset{o}{A}$.

12. $m_1V_1^2 = m_2V_2^2 = 3KT$
$= \frac{3RT}{N_A} = \frac{3PM}{N_A d}$

13. Angular momentum conservation about axel of stool.
$I_\omega w_s = I_{sp}w_f + I_\omega (w_f - w_s) + m_\omega d^2 wf$
$w_f = \frac{3}{8}w_s$

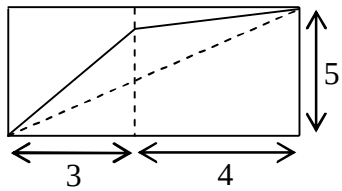
14. Linear impulse $= \Delta p = mv_{cm} - 0$
$= \frac{3}{8}m_\omega d w_s$

15. Slope will be now different such that
$\sin\theta = \frac{200}{500} = \frac{2}{5}$
So, $\cos\theta = \sqrt{\frac{21}{25}}$
Hence, T = $\frac{2u\sin 30^o}{g\cos\theta}$

16. For minimum slope, line will be PQ, i.e. a horizontal line
Hence, time = $\frac{2u\sin 30^o}{g}$ = 1 sec.

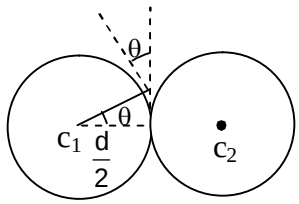
17. $\sqrt{7^2+5^2} = \sqrt{74}$

18. Snell's law
For minimum time it will follow a path which could have been followed by a light ray under refraction, but not the previous one. Definitely time taken would be less than what could have been over straight path.



SECTION – C

1. $4T \cdot \cos\theta \cdot dl = 2T \cdot dl$
$\cos\theta = \frac{1}{2}$
$\theta = 60^o$
So, $\frac{d}{2} = R\cos 60 = \frac{R}{2}$
c_1c_2 = d = R = 2 m



2. $f_n = \dfrac{1}{\dfrac{2\pi r_n}{v_n}} = \dfrac{v_n}{2\pi r_n}$

$f_1 = \dfrac{2.18\times10^6}{2\times3.14\times0.529\times10^{-10}}\ \sec^{-1}$

$= 6.56\times10^{15}\ \sec^{-1}$

$P = 6.56$

$\dfrac{2P}{13} = 1$ approximately.

3. $dQ = 2\pi x dx.\dfrac{Q}{\pi R^2}$, $E_x = \dfrac{-x}{2}\dfrac{2B}{t}$

$\Delta L = \int \tau\ dt \Rightarrow IW = \int_0^R \dfrac{x}{2}\dfrac{2B}{t}\ .\ 2\pi x dx \dfrac{Q}{\pi R^2} dt$

$\dfrac{MR^2}{2}\omega = \dfrac{QBR^2}{2} \Rightarrow \omega = \dfrac{QB}{M} = 4$

4. $V_{avg} = \dfrac{v_1 + v_2}{2}$ Since variation is linear, $t = \dfrac{t}{V_{avg}} = 2\sqrt{\dfrac{M}{\gamma R}}$

$= \sqrt{\dfrac{\gamma R}{M}}\left(\dfrac{\sqrt{T_1}+\sqrt{T_2}}{2}\right)$

5. $\dfrac{1}{f}+\dfrac{1}{f} = 30$, $f = 60$

$\dfrac{1}{v_1}-\dfrac{1}{-20} = \dfrac{1}{60} \Rightarrow v_1 = -30$, $\dfrac{1}{v}-\dfrac{1}{-30} = \dfrac{1}{60}$

$\dfrac{I}{0} = m = \dfrac{v}{v_1} = \dfrac{-60}{-30} = 2$

$\Rightarrow \dfrac{I}{-5} = 2 \Rightarrow I = -10$

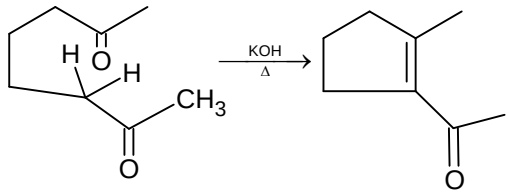
y coordinate = -5

Chemistry

PART – II

SECTION – A

1. Normally not possible to F_2 because it is highly exothermic with I_2 reversible reaction takes place due to more E.N of Cl_2 compare to Br_2. It does not give e^- pair to carbocation.

2. 

3. II, III are unstable because of their rigid frame work, to stabilize themselves by collapsing to the stable intermediate planer state. II is more stable than III because the greater rigidity about bridge head carbon with a one-carbon than with a two – carbon bridge.

4. $5e^- + Mn^{7+} \longrightarrow Mn^{2+}$

$\left(P^{3+}\right)_4 \longrightarrow 4P^{5+} + 8e^-$

Meq. Of $KMnO_4 = \dfrac{100 \times 5 \times 1000}{158} = 3164.56$

Meq. Of $P_4O_6 = \dfrac{100 \times 8 \times 1000}{219.9} = 3638.02$

Meq. Of P_4O_6 in excess = 473.46

$\therefore \dfrac{w \times 8 \times 1000}{219.9} = 473.46 \quad W_{P_4O_6}$ in excess $= 13.01$g

5. $NH_4CN(s) \rightleftharpoons NH_3(g) + HCN(g)$

P P

At eq.

$\therefore$ Total pressure = 2P = 0.298 atm

p = 0.149

Also $K_P = P^1_{NH_3} \times P^1_{HCN} = 0.149 \times 0.149 = 0.022$ atm^2

Now $P_{NH_3} = 0.25$ atm

$NH_4CN(s) \rightleftharpoons NH_3(g) + HCN(g)$

Initial	0.25	0
At eq	$0.25 + P^1$	P^1

$K_{P = P^1}$ $(0.25 + P^1)$

$P^1 = 0.0694$ atm

6. $AgBr(s) \rightleftharpoons Ag^+(aq.) + Br^-(aq.) \qquad K_1 = K_{sp}$

$Ag^+(aq) + 2S_2O_3^{2-}(aq.) \rightleftharpoons Ag(S_2O_3)_2^{3-}(aq.) \qquad K_2 = K_f$

$AgBr + 2S_2O_3^{2-}(aq.) \rightleftharpoons Ag(S_2O_3)_2^{3-} + Br^- \qquad K = K_{sp} \times K_f = 25$

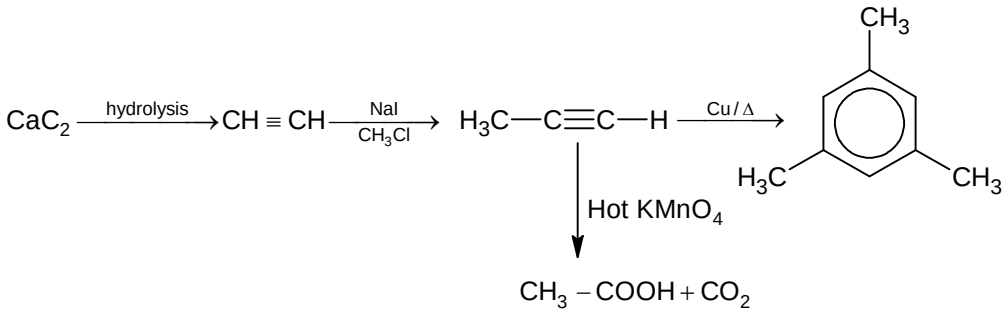
$$K = \frac{x^2}{(0.1-2x)^2} \Rightarrow \frac{x^2}{(0.1-2x)^2} = 25$$
$$\Rightarrow x = 0.045\ M.$$

8. $M \longrightarrow M^+ + e^-$

$$M^+ + e^- + (x+y)NH_3 \longrightarrow \underset{\text{solvated metal}}{M(NH_3)_x} + \left[\underset{\text{solvated } e^-}{e^-(NH_3)_y}\right]$$
$$\downarrow \text{on stand}$$
$$M - NH_2$$

11. V. B.T does not explain colour, Cu^{2+}, Pt^{2+}, Pd^{2+} square planar geometries with proper reason.

14.



15. $W = -P.\Delta V = -nR\Delta T = -2 \times 8.314 \times 600 = -9.97\ kJ$

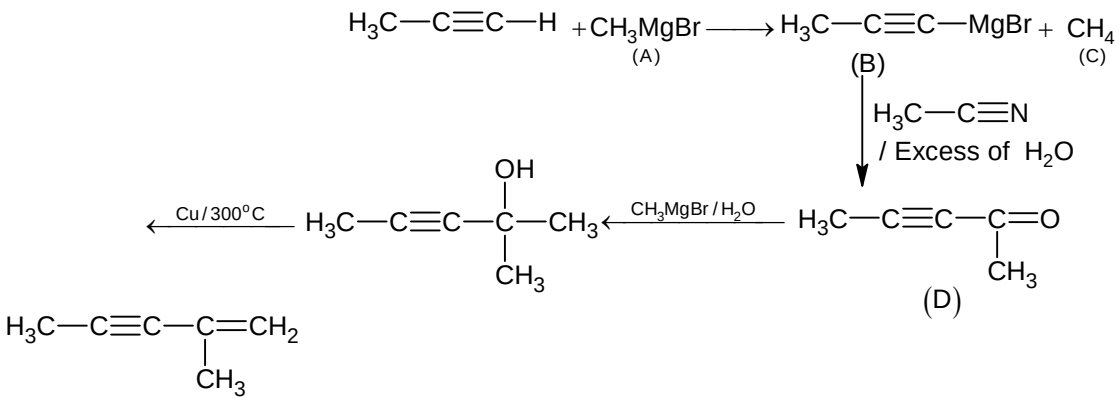
16. $\Delta H_{Total} = \Delta H_{AB} + \Delta H_{BC}$
$= nC_{p,m}\Delta T + 0$
$= 2 \times \frac{7}{2} \times R \times (800-200)$
$= 4200\ R$

17. Due to high electronegativity of 'F' bp–bp repulsion is minimum in OSF_2. Steric effect may also be the explanation.

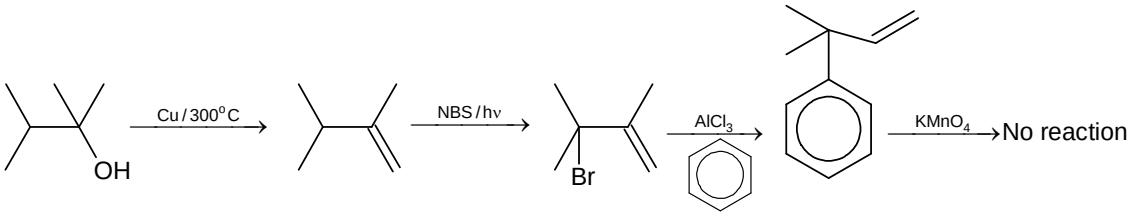
18. Among P, As and Sb, P is most electronegative and so it exerts the strongest pull on shared electrons causing more bp–bp repulsion.

SECTION – C

1.



2.



Due to absence of Benzoic hydrogens.

3. $NH_4NO_3 \xrightarrow{\Delta} \underset{(A)}{N_2O} + 2H_2O$

$N_2O \xrightarrow{1300^oC} \underset{(B)}{2NO} + \underset{(C)}{N_2}$

$NO + FeSO_4 \longrightarrow \underset{\text{Brown Compound}}{\underset{(D)}{FeSO_4.NO}}$

5. $\frac{P^o - P}{P} = \frac{n}{N} = \frac{W}{W \times N}$

$= \frac{850 - 844.9}{844.9} = \frac{2 \times 76}{M \times 100}$

$M \approx 252$

$N = \frac{252}{32} \approx 8$

Mathematics

PART – III

SECTION – A

1. Consider $f(x)=\int_0^x\left(1+\cos^8 x\right)\left(ax^2+bx+c\right)dx$

Obviously, f(x) is continuous in [1, 2] and differentiable in (1, 2). Also f(1) = f(2)
By Rolle's theorem, there exist one point
$k \in (1, 2)$ such that $f'(k) = 0$ ..... (1)
Now, $f'(k) = (1 + \cos^8 k)(ak^2 + bk + c) = 0$
$\Rightarrow ak^2 + bk + c = 0$ $(1 + \cos^8 k \neq 0)$
$\therefore x = k$ is root of $ax^2 + bx + c = 0$ where $k \in (1, 2)$
Here, atleast one root $\in (1, 2)$

2. $\because$ $2\cos 2x - 1 = 2(\cos^2 x - \sin^2 x) - (\sin^2 x + \cos^2 x) = \cos^2 x - 3\sin^2 x$
The given equation reduces to $b \sin x = b + \sin x$, then

$$\sin x = \frac{b}{b-1}$$

$$\Rightarrow -1 \le \sin x \le 1$$

$$\Rightarrow -1 \le \frac{b}{b-1} \le 1$$

$$\Rightarrow \frac{b}{b-1} \ge -1$$

and $$\frac{b}{b-1} \le 1$$

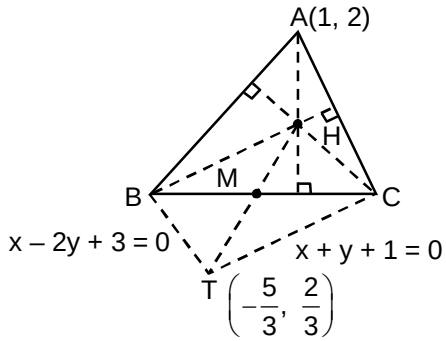
$$\Rightarrow b < \frac{1}{2}$$

$$\Rightarrow b \in \left(-\infty, \frac{1}{2}\right)$$

3. The vectors representing forces are 2 × unit vector along $(2\hat{i} + 4\hat{j} + 4\hat{k})$
and 3 × unit vector along $(4\hat{i} - 4\hat{j} + 2\hat{k})$
i.e., $\frac{1}{3}(2\hat{i} + 4\hat{j} + 4\hat{k})$ and $\frac{1}{2}(4\hat{i} - 4\hat{j} + 2\hat{k})$
$\therefore$ The resultant is $\frac{1}{3}(8\hat{i} - 2\hat{j} + 7\hat{k})$
The displacement is given by the vector $\hat{i} + 2\hat{j} + \hat{k}$
Hence the work done $= \frac{1}{3}(8\hat{i} - 2\hat{j} + 7\hat{k}) \cdot (\hat{i} + 2\hat{j} + 2\hat{k}) = 6$ gm-cm

4. Let r be the common ration of the given G.P., then $b = ar$, $c = ar^2$, and $d = ar^3$
Then $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2)$
$= (a^2 + a^2r^2 + a^2r^4)(a^2r^2 + a^2r^4) + a^2r^6$
$= a^2(1 + r^2 + r^4)\, a^2r^2(1 + r^2 + r^4)$
$= a^4r^2(1 + r^2 + r^4)^2$
$= (a^2r + a^2r^3 + a^2r^5)^2$
$= (a \cdot ar + ar \cdot ar^2 + ar^2 \cdot ar^3)^2$
$= (ab + bc + cd)^2$

8. M is mid point of BC and HT
$\therefore$ BTCH is a parallelogram
$\because$ CH is perpendicular BA and CH is parallel BT
$\therefore$ BT is perpendicular AB
Similarly CT is perpendicular AC
Now, a circle drawn taking AT as diameter will pass through B and C
i.e. points A, B, T, C are concyclic points
$\therefore$ Required circle $(x-1)\left(x+\frac{5}{8}\right)+(y-2)\left(y-\frac{2}{3}\right)=0$



12. First use integrating by parts to show that
$u_n = \int_0^{\pi/2} \cos^{n-1}x \sin x \sin nx \, dx$
$= \int_0^{\pi/2} \cos^{n-1}x \cos(n-1)x - \int_0^{\pi/2} \cos^{n}x \cos nx \, dx$ (Use sin nx sin x = cos(n – 1)x – cos nx cos x)
$= u_{n-1} - u_n$
$\Rightarrow \frac{u_n}{u_{n-1}} = \frac{1}{2}$
We have, $u_1 = \frac{\pi}{4}$, $u_n = \frac{1}{2^{n-1}} \cdot \frac{\pi}{4} = \frac{\pi}{2^{n+1}}$

13.-14. Let α be the common root of the three equations are their other roots be β, γ, δ respectively
$\therefore \alpha + \beta = -a$, $\alpha\beta = 12$
$\alpha + \gamma = -b$, $\alpha\gamma = 15$
$\alpha + \delta = -(a + b)$, $\alpha\delta = 36$
$\therefore (\alpha + \beta) + (\alpha + \gamma) = -(a + b) = \alpha + \delta$
$\therefore \alpha + \beta + \gamma = \delta$ ..... (1)
Again $\alpha(\beta + \gamma + \gamma) = 12 + 15 + 36 = 63$
or $\alpha(2\delta - \alpha) = 63$ by (1)
or $2\alpha\delta = \alpha^2 = 63$ or $72 - \alpha^2 = 63$
$\because \alpha\delta = 36$
$\therefore \alpha^2 = 9$; $\alpha = 3, -3$
$\alpha = 3 \Rightarrow \beta = 4$, $\gamma = 5$, $\delta = 12$
$\alpha = -3 \Rightarrow \beta = -4$, $\gamma = -5$, $\delta = -12$
$\therefore a = -(\alpha + \beta) = 7, -7$; $b = -(\alpha + \gamma) = 8, -8$

15. Equation of any tangent to the parabola $y^2 = 8x$ is $y = mx + \frac{2}{m}$
Since, it touches $\frac{x^2}{4} + \frac{y^2}{15} = 1$ ($\because c^2 = a^2m^2 + b^2$)
$\left(\frac{2}{m}\right)^2 = 4m^2 + 15 \Rightarrow m = \pm\frac{1}{2}$
Hence, equations of tangent are $x \pm 2y + 8 = 0$

16. When $m = \frac{1}{2}$ then slope of normal is –2 and equation of the normal to the parabola is
$y = -2x - 2(2)(-2) - 2(-2)^3$ ($\because y = mx - 2am - am^3$)
$\Rightarrow y = -2x + 8 + 16 \Rightarrow 2x + y = 24$

SECTION – C

1. $$\frac{\cos x(1+4\cos 2x)}{\sin x(1+4\cos^2 x)} = \frac{\cos x\left[1+4(1-2\sin^2 x)\right]}{\sin x\left[1+4-4\sin^2 x\right]} = \frac{\cos x(5-8\sin^2 x)}{\sin x(5-4\sin^2 x)} = \frac{\cos x}{\sin x} - \frac{4\sin x}{(5-4\sin^2 x)}$$

Put t = sin x; dt = cos x dx

2. $$\int_0^{x^2} f(t)\,dt = x\cos\pi x$$

$$2x\cdot f(x^2) = \cos\pi x - \pi x\sin\pi x$$

Put x = 2

$4f(4) = 1$

$$f(4) = \frac{1}{4}$$

And f(x), $\int_0^{f(x)} t^2 dt = x\cos\pi x$

$\{f(x)^2\cdot f'(x) = \cos\pi x - \pi x \sin\pi x$

Put x = 4

3. $f''(x) - 5f'(x) + 6f(x) \geq 0 \Rightarrow f''(x) - 2f'(x) \geq 3(f'(x) - 2f(x))$

$\Rightarrow g'(x) \geq 3g(x)$; where $g(x) = f'(x) - 2f(x)$

Multiply by e^{-3x} and then we can write it as

$$\Rightarrow \frac{d}{dx}\left(g(x)e^{-3x}\right) \geq 0 \Rightarrow y = g(x)e^{-3x} \text{ is non decreasing}$$

$\because x \geq 0$ (Given)

$\Rightarrow g(x)e^{-3x} \geq g(0)\,e^{-0} \Rightarrow g(x)\,e^{-3x} \geq -2 \Rightarrow g(x) \geq -2e^{3x}$

$\because g(x) = f'(x) - 2f(x)$, $g(0) = f(0) - 2f(0) = 0 - 2$

$\Rightarrow f'(x) - 2f(x) \geq -2e^{3x}$

Multiply it by e^{-2x} and than we can write it as

$$\Rightarrow \frac{d}{dx}\left(f(x)e^{-2x} + 2e^{x}\right) \geq 0 \Rightarrow f(x)\,e^{-2x} + 2e^{x} \geq 3$$

$\because x \geq 0$, $e^x \geq 1$ ..... (1)

$\Rightarrow$ We have $e^{-2x} f(x) + 2e^x \geq 3$

or $f(x) \geq 3e^{2x} - 2e^{3x}$ ..... (2)

$f(x) \geq ah(bx) - bh(ax)$ ..... (3)

From equation (2) and (3), we get a = 3, b = 2

$\therefore (a + b)\,h(0) = (3 + 2)\,e^0 = 5$

4. $$\frac{dy}{dx} + y = 4xe^{-x}\sin 2x$$

Integrating factor = e^x

$$\therefore \text{Equation becomes } ye^x = 4\int x\sin 2x\,dx$$

$\Rightarrow ye^x = (\sin 2x - 2x\cos 2x) + c$

$\because f(0) = 0 \Rightarrow c = 0$

$\therefore y = f(x) = (\sin 2x - 2x\cos 2x)e^{-x}$

$f(k\pi) = e^{-k\pi}(0 - 2k\pi(1))$

$f(k\pi) = -2k\pi\, e^{-k\pi}$

$$\lim_{n\to\infty}\sum_{k=1}^{n} f(k\pi) = -2\pi\left(1\cdot e^{-\pi} + 2e^{-2\pi} + 3e^{-3\pi} + \ldots.. \text{upto } \infty\right) = \frac{-2\pi e^{\pi}}{\left(e^{\pi}-1\right)^2}$$